GLASS HEROES

Business Manager — v0.3.3.0 RC1 Developer Guide

GTA V Legacy (1.0.3889.0) • ScriptHookVDotNet API 2.11.6 • .NET Framework 4.8

Release status and evidence

- Experimental source candidate. Compiled: NO (C# / GTA references unavailable here). In-game tested: NO.
- Last successful user-tested version: v0.3.2.9a with Los Santos RED and approximately 73 active scripts/plugins; manager hiring and wholesale pickup/delivery succeeded.
- Crash-free in that session does not yet prove repeated cold-start reliability. Preserve the v0.3.2.9a rollback copy.
- The candidate retains the previous boot gate and HoldGameplayTick diagnostics; default startup delay is 30 seconds.

Release objective

Address cargo floating above ground at the wholesale warehouse and in the shop, make manager walking configurable like the receptionist, repair placement menu feedback, and add safe supplier-order recovery/cancellation without compromising the startup-stability baseline.

Roadmap to v1.0

V4: tablet and phone, SHVDN3 migration. V5: missions. V6: employee interactions/skills. V7: realistic tablet parts ordering. V8–V9: optimization, fixes, performance. V1.0: stable release. Cold-start stability stays the immediate priority.

Implementation inventory

ID	Subsystem	Change & reason
GH-033-01	Warehouse visuals	Spawn a local ground-placed pallet and anchor the physical parcel above it; clean up when distant or collected.
GH-033-02	Shop storage visuals	Base stacked parcel positions on actual pallet prop position rather than potentially elevated saved Z; delay handoff until pallet exists.
GH-033-03	Manager walking	Independent DIRECT/CUSTOM 0–5 waypoint route, optional final approach, clipboard scenario and bounded stalled-route retries.
GH-033-04	Menu hierarchy	Manager profile structured like receptionist; manager workstation and approach moved into employee placement settings.
GH-033-05	Placement UI	Point saving enables marker previews; color markers higher and thicker;

		explicit radius/shop-location notices.
GH-033-06	Order recovery	Reset existing pending shipment and owned visuals without issuing invoice, inventory or duplicate order.
GH-033-07	Order cancellation	Double-press confirmation within 12s; remove pending record and owned visuals; refund legacy prepaid business cash.
GH-033-08	Supplier ETA	Use active unpaused Game.GameTime intervals and persist remaining gameplay seconds every 15 active seconds.
GH-033-09	Financial integrity	Make manager-training price display a pure calculation; no savings credited from merely opening menu.
GH-033-10	Startup	Carry forward diagnostic gating unchanged; do not add constructor-time native calls.

Saved data & compatibility

Additive keys: [EmployeeSettings] CustomManagerPaths; [ManagerApproach] Set/X/Y/Z/Heading; [ManagerWaypoint1] ... [ManagerWaypoint5] Set/X/Y/Z/Heading; [Purchasing] DeliveryRemainingGameSeconds. Existing locations, personnel, orders, parts, cash, wages, and configuration keys are retained. No Save.ini reset or forced migration is required.

Legacy pending delivery migration: when the new remaining-seconds key is missing, the delivery restarts with its configured supplier lead time. This conservatively prevents old real-world elapsed time from completing an order while GTA was closed. DeliveryDueUtcTicks remains for legacy identity/compatibility and is not used to advance ETA.

Full before/after excerpts (representative; full patch is included)

01 — Warehouse crate grounding

The old package uses a fixed world point and may hover when the local floor differs. The updated code creates a pallet with placeOnGround=true, then positions the crate relative to that pallet. A warehouse box cannot be newly collected until its pallet model exists.

BEFORE

```
_supplierCrate = CreateFixedEquipmentProp("prop_cs_cardbox_01",
  _wholesaleWarehousePickup.Position,
  _wholesaleWarehousePickup.Heading, false);
```

AFTER

```
UpdateWholesalePalletVisual();
if (_wholesaleWarehousePallet != null && _wholesaleWarehousePallet.Exists())
  _supplierCrate = CreateFixedEquipmentProp("prop_cs_cardbox_01",
    WarehousePackagePosition(), _wholesaleWarehousePickup.Heading, false);
```

02 — Shop pallet grounding

Stacked boxes now inherit the same grounded pallet Z. Physical pallet loading is checked before the final stockroom box is detached and frozen.

BEFORE

```
return pallet.Position + new Vector3(localX * cos - localY * sin,
    localX * sin + localY * cos, 0.28f + layer * 0.37f);
```

AFTER

```
Vector3 palletGround = _storagePalletProps[palletSlot] != null &&
    _storagePalletProps[palletSlot].Exists() ?
        _storagePalletProps[palletSlot].Position : pallet.Position;
return palletGround + new Vector3(localX * cos - localY * sin,
    localX * sin + localY * cos, 0.28f + layer * 0.37f);
```

03 — Manager navigation and behavior

The manager previously issued direct tasks toward the workstation. Now the manager can follow 0–5 optional MLO stops and a final approach; DIRECT mode uses navmesh. The in-world route advances by proximity and limits retry requests instead of issuing infinite tasks.

BEFORE

```
GTA.Native.Function.Call(GTA.Native.Hash.TASK_GO_STRAIGHT_TO_COORD,
    _managerWorker.Handle, post.Position.X, post.Position.Y, post.Position.Z,
    1f, -1, post.Heading, 0.5f);
```

AFTER

```
foreach (WorkBay stop in _managerRoute)
    if (stop.IsSet) _managerPathNodes.Add(stop.Position);
if (_managerApproach.IsSet) _managerPathNodes.Add(_managerApproach.Position);
_managerPathNodes.Add(post.Position);
WalkManagerRouteNode(now);
```

04 — Supplier ETA no longer wall-clock based

An order may now advance only while the unpaused game is actively ticking. The remaining duration is persisted and will not catch up across GTA shutdown.

BEFORE

```
_deliveryDueUtcTicks = DateTime.UtcNow.AddSeconds(
    SupplierDeliverySeconds(_selectedSupplier)).Ticks;
if (_deliveryDueUtcTicks > now.Ticks) return;
```

AFTER

```
_deliveryRemainingGameSeconds = SupplierDeliverySeconds(_selectedSupplier);
// OnTick, only when activeGameSeconds > 0:
_deliveryRemainingGameSeconds = Math.Max(0,
    _deliveryRemainingGameSeconds - activeGameSeconds);
if (_deliveryRemainingGameSeconds > 0) return;
```

05 — Manager training-price side effect removed

Prevents savings from growing simply because a menu redraw evaluates the displayed price. Savings are recorded only when training is actually paid for.

BEFORE

```
return "$" + ManagerAdjustedExpense(750).ToString("N0");
// ManagerAdjustedExpense() modifies _managerSavings.
```

AFTER

```
return "$" + (750L * (100 - ManagerDiscountPercent()) /
    100).ToString("N0");
// No mutation from rendering the menu.
```

06 — Order recovery/cancel actions

Recovery is idempotent with respect to financial and inventory records, while cancellation avoids accidental abandonment of a legitimate shipment. For exact source see full unified diff.

BEFORE

```
else if (_selection == 9) ShowStatusNotice("SUPPLIER DELIVERY ROUTE",
```

AFTER

```
else if (_selection == 10) RecoverSupplierShipment();
else if (_selection == 11) CancelSupplierOrderWithConfirmation();
```

07 — Marker visibility and manager placement

Brings manager physical positions alongside other employees. Point Set/Update enables the saved-point preview and raises rendered markers to reduce floor occlusion; marker display still requires a saved shop and the player within 80 m.

BEFORE

```
return new string[] { "Staff Break Area >", "Mechanic Workstations >",
  "Receptionist Desk >", "Receptionist Desk Approach >",
  "Staff Spawn / Entry >", "< Back to Employee Settings" };
```

AFTER

```
return new string[] { "Staff Break Area >", "Mechanic Workstations >",
  "Receptionist Desk >", "Receptionist Desk Approach >",
  "Manager Workstation >", "Manager Workstation Approach >",
  "Staff Spawn / Entry >", "< Back to Employee Settings" };
```

Installation / upgrade / rollback

- Make three separate backups: scripts/GlassHeroesBusiness.2.cs, scripts/GlassHeroesBusiness/Save.ini, and scripts/GlassHeroesBusiness/Settings.ini.
- Copy only Install/scripts/GlassHeroesBusiness.2.cs from ZIP into GTA V/scripts. Do not install older copies, duplicate DLLs, or a second .2.cs.
- Do not overwrite Save.ini or Settings.ini. Keep [Diagnostics] HoldGameplayTick=false and StartupDelaySeconds=30 as successful baseline.
- Cold-start into unpaused Story Mode; wait at least 45 seconds before trying F2. Do not use the script/plugin controller for this first test.
- If the cold start fails, restore the exact v0.3.2.9a source; supply fresh ScriptHookVDotNet.log and scripts/GlassHeroesBusiness/Errors.log. Leave save files untouched.

Exact verification performed

Test	Result
Source brace, parenthesis, bracket balance + ordering	PASS
MenuPage references, names and duplicate labels	PASS
No duplicate detected method names or switch cases	PASS
Manager route/load/save keys and order paths present	PASS
Unpaused game-time ETA and unchanged boot gating	PASS
No manager-menu savings mutation	PASS
Static checks total	21 / 21 PASS

C# compilation	NOT RUN — C# toolchain / game references unavailable
GTA V cold-start, active 73-script test, keyboard controls	NOT RUN — requires user installation

Required in-game regression tests

☐ Run at least three complete GTA cold starts with RAGE Plugin Hook disabled first. Then compare with LS RED and your standard ~73-script setup.

☐ Verify F2 and keyboard Enter/Up/Down one activation per press. Inspect employee placement point status before/after Set and Clear.

☐ Set shop and all placement points while closed; enable preview within 80 m; confirm no invisible or clipping saved markers.

☐ Hire manager, set workstation and optional manager approach, choose CUSTOM walking, set 1–5 stops, open shop and watch single manager walk each stop to clipboard scenario.

☐ Leave streaming distance and return; look for one manager only, no orphaned peds or repeating task faults.

☐ Order WHOLESALE, assign vehicle and wait in active GTA until ETA 0. At fixed warehouse, verify pallet rests on ground and box rests on pallet; pick up, load, drive, unload, and stock.

☐ At shop, verify each of 1–4 purchased pallets and cargo boxes rest on grounded pallet; repeat with an MLO raised floor.

☐ Start a fresh wholesale order, choose RECOVER; ensure original order persists, no inventory credited and no duplicate invoice. Collect again.

☐ Start a pending order, press CANCEL once (no action), press again within 12 seconds (cancels). Verify inventory and invoice unchanged. Test legacy prepaid refund separately on backed-up save.

☐ Place pending order; exit GTA and return after real-world time. Supplier ETA must not progress while game is closed. Pause GTA and confirm ETA does not progress.

☐ Check business cash, wages, parts, stocked crates and completed orders after save/reload; look for no duplication or drops. Review Errors.log.

Known issues and limits

- Not compiled or run in GTA here. Behavior under 73 scripts is unverified for this candidate; repeated cold-start stress tests are mandatory.
- Warehouse pickup point is fixed at (-164.2800, -2676.5900, 6.0141). Ground placement uses GTA prop placement and may still need a venue-specific Z adjustment on altered collision/MLOs.
- The stored pallet prop model is assumed available; handoff now waits when the prop has not loaded. It retries rather than inventing a stock credit.
- Manager peds intentionally avoid visible or very close spawn points to reduce pop-in. A manager can remain visually absent until the entrance is off camera. Stalled MLO waypoints do not forcibly teleport the manager.
- New orders are billed on stocking. Canceling a legacy prepaid pending order refunds the saved order cost to cash, but older historical expenditure summaries may still include the original payment.
- Delivered packages have a generic rear-cabin vehicle offset that may need model-specific tuning. Neither order recovery nor pallet changes resolve collision inside every modded vehicle.